

# Revisiting Communication in Federated Learning: Parameter Vs. Knowledge-Based Strategies

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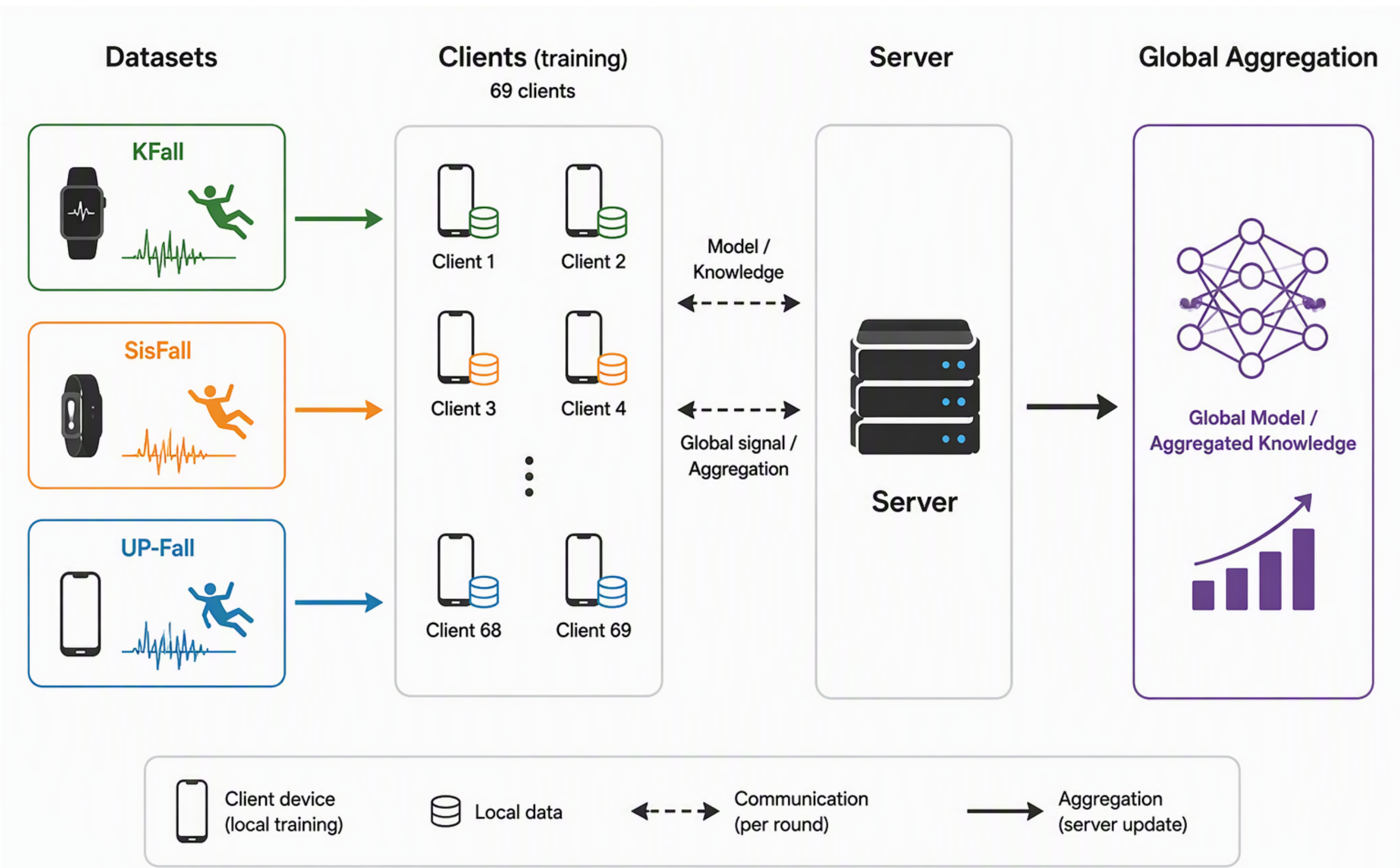
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## Motivation & Research Goals

Federated Learning (FL) enables collaborative model training without sharing raw data, making it particularly suitable for privacy-sensitive healthcare applications. However, performance strongly depends on the communication mechanism used between clients and the server, especially under heterogeneous and non-IID data distributions. This work compares parameter-sharing and knowledge-based communication strategies for wearable fall detection, investigating their impact on global performance, personalization capability, and communication efficiency.

## Methods



Datasets: KFall, SisFall and UP-Fall  
Experimental Setup: 69 Training Clients, 9 Development Clients and 9 Test Clients  
Model : 32 → 256 → 128 → 64 → 1

| Strategy            | Transferred Information      |
|---------------------|------------------------------|
| FedAvg              | Model weights                |
| KD + Weight Sharing | Model weights + distillation |
| Soft Labels         | Prediction statistics        |
| Class Prototypes    | Class-level representations  |

## Key References

[1] McMahan et al. (2017), Communication-Efficient Learning of Deep Networks from Decentralized Data  
[2] Li et al. (2020), Federated Learning: Challenges, Methods, and Future Directions  
[3] Kairouz et al. (2021), Advances and Open Problems in Federated Learning  
[4] Li & Wang (2019), FedMD: Heterogeneous Federated Learning via Model Distillation  
[5] Wu et al. (2022), Communication-Efficient Federated Learning via Knowledge Distillation

For the complete paper, supplementary material, and full bibliography, please scan the QR code.



## Selected Results

| Global vs Local Performance |               |              |
|-----------------------------|---------------|--------------|
| Strategy                    | Global PR-AUC | Local PR-AUC |
| FedAvg                      | 0.969         | 0.915        |
| KD + Weight Sharing         | 0.957         | 0.914        |
| Soft Labels                 | 0.732         | 0.809        |
| Class Prototypes            | 0.701         | 0.792        |

| Communication Cost  |                  |
|---------------------|------------------|
| Strategy            | Values Exchanged |
| FedAvg              | 49,665           |
| KD + Weight Sharing | 49,665           |
| Class Prototypes    | 132              |
| Soft Labels         | 2                |

**Main Finding**  
FedAvg achieved the highest global performance (PR-AUC = 0.969). However, knowledge-based communication strategies substantially reduced communication costs while maintaining competitive local performance. Soft-label communication exchanged only 2 values per round and achieved a local PR-AUC of 0.809.

## Conclusion

FedAvg achieved the best global performance under heterogeneous non-IID data. In contrast, knowledge-based communication strategies showed lower global performance but remained competitive at the local level while dramatically reducing the amount of information exchanged. These findings suggest that knowledge transfer is a promising alternative for personalization-oriented Federated Learning, where communication efficiency and client-specific performance are primary objectives.

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